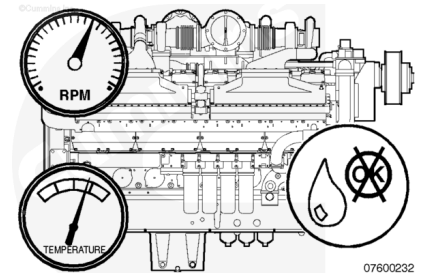


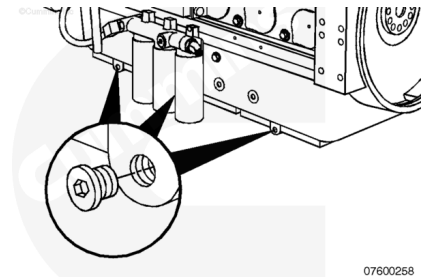
Trainings ▾

Drain

Operate the engine until the water temperature reaches 60°C [140°F].

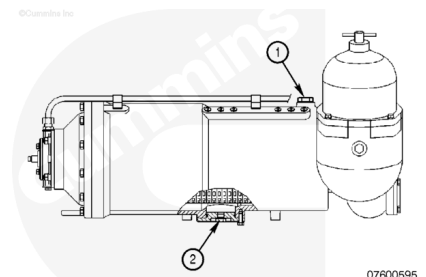


The QSK45 and QSK60 engines have various lubricating oil pan configurations. Remove all drain plugs when draining the oil pan.



⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

Remove the oil drain plugs and drain the oil immediately to make sure all the suspended contaminants are removed from the engine.

For engines with the Eliminator™ filter system, also drain the oil from the Eliminator™ by removing the vent plug (1) and the oil drain plug (2).

Fill

Use a high quality 15W-40 multiviscosity oil that meets the American Petroleum Institute (API) performance classification CE or CF4 or better.

Choose the correct oil for the operating climate. Refer to Procedure 018-003

(/qs3/pubsys2/xml/en/procedures/102/102-018-003.html) in the Operation and Maintenance Manual, QSK45 and QSK60 Series Engines, Bulletin 3666260 correct oil.



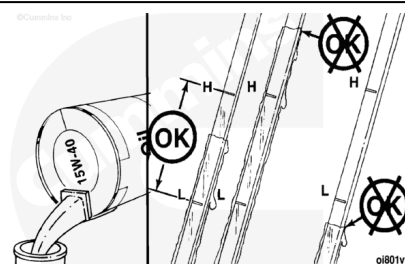
Fill the engine to the correct level with clean engine oil.

Refer to Procedure 018-017

(/qs3/pubsys2/xml/en/procedures/56/56-018-017.html) for engine sump capacity.

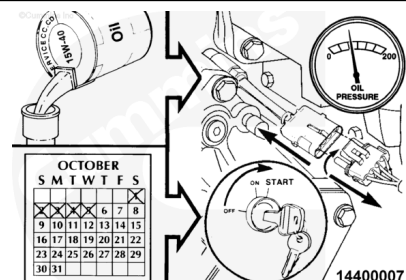
Check the oil level on the dipstick. The level **must** be to the high (H) mark on the dipstick.

Do **not** overfill. Drain off excess oil in a pan.



⚠ CAUTION ⚠

Before starting engines not equipped with a PRELUB™ system, complete the following steps. Engine damage will occur because of inadequate lubrication at startup if these steps are not followed.



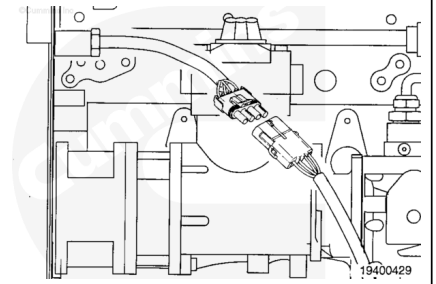
Note : Engines that are equipped with a PRELUB™ system are self priming and do not require completion of this and the next step.

The following steps **must** be completed after each oil change or after the engine has been shutdown for more than five days to make sure the engine receives the correct oil flow through the lubricating system.

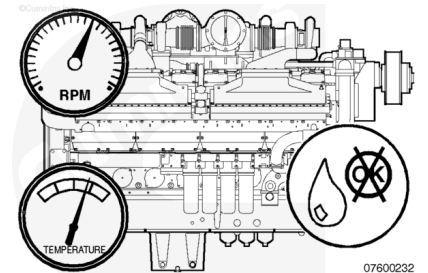
Disconnect the engine position sensor.

Using the starter motor, rotate the engine crankshaft until oil pressure appears on the gauge or the warning light goes out.

Connect the engine position sensor.



Operate the engine at idle speed to inspect for leaks at the filters and drain plug(s).



Shut off the engine.

Wait approximately 5 minutes to allow the oil to drain from the upper parts of the engine.

Check the oil level.

Add oil as necessary to bring the oil level to the high (H) mark on the dipstick.

